

COMPSCI 1DM3 - Winter 2026

Discrete Mathematics for Computer Science

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Last Lecture:

- Summation:

$$\sum_{j=0}^n a_j = \sum_{j=\textcolor{red}{1}}^{\textcolor{red}{n+1}} a_{j-\textcolor{blue}{1}} = \sum_{j=\textcolor{red}{2}}^{\textcolor{red}{n+2}} a_{j-\textcolor{blue}{2}} = \sum_{j=\textcolor{red}{m}}^{\textcolor{red}{n+m}} a_{j-\textcolor{blue}{m}}$$

- **Geometric progression.** For the sequence $\{a_k\}$ with $a_k = a \cdot r^k$ we have

$$\sum_{k=0}^n ar^k = \begin{cases} \frac{a(r^{n+1} - 1)}{r - 1} & \text{if } r \neq 1 \\ (n+1)a & \text{if } r = 1 \end{cases}$$

- **Telescopic sum.** Given a sequence $\{a_j\}$, we have

$$\sum_{j=1}^n (a_j - a_{j-1}) = a_n - a_0.$$

Double sum

Suppose we have a sequence with two indices: $\{a_{i,j}\}_{i,j \in \mathbb{Z}}$. How do we compute the sum of its terms?

$$\sum_i \sum_j a_{i,j}$$

Example

Compute $\sum_{i=0}^3 \sum_{j=0}^2 (3i + 2j)$.

$$\begin{aligned} & (3i + 2 \cdot 0) + (3i + 2 \cdot 1) + (3i + 2 \cdot 2) \\ &= 9i + 6 \end{aligned}$$

$$\begin{aligned} & \hookrightarrow \sum_{i=0}^3 (9i + 6) = (9 \cdot 0 + 6) + (9 \cdot 1 + 6) + (9 \cdot 2 + 6) + (9 \cdot 3 + 6) \\ &= \end{aligned}$$

Sec 2.5: Cardinality of sets

- In Sec 2.1, we called sets with finite cardinality as *finite* sets.
- In this section, we like to briefly study cardinality of infinite sets.
- In particular, what it means for two infinite sets to have equal cardinality.
- To do so, we use mappings between sets: injections and bijections

Cardinality of sets

Definition

The sets A and B have the same cardinality if and only if there is a bijection

$$\underline{f} : \underline{A} \rightarrow \underline{B}.$$

Definition

- If there is an injection $A \hookrightarrow B$, we say that the cardinality of A is less than or the same as the cardinality of B and write $|A| \leq |B|$.
- If $|A| \leq |B|$ and A and B have different cardinality, then the inequality is strict: $|A| < |B|$.

Countable sets

Definition

- A set A is *countable* if either A is finite or there is a bijection $f : A \rightarrow \mathbb{Z}^+$.
- A set that is not countable is called *uncountable*
- The cardinality of an uncountable set is denoted by $\aleph_0$

Equivalently:

A set A is countable if its elements can be listed in a sequence indexed by $\mathbb{Z}^+$:

$$a_1, a_2, a_3, \dots, a_n, \dots$$


Example

$$\rightarrow |O| = |\mathbb{Z}^+|$$

- The set of odd positive integers is a countable set.

$$f: O \rightarrow \mathbb{Z}^+, f(n) = 2n - 1$$

injective:

$$\text{Suppose } f(n_1) = f(n_2), 2n_1 - 1 = 2n_2 - 1 \\ n_1 = n_2 \checkmark$$

- Is $\mathbb{Z}$ countable? In other words, is $|\mathbb{Z}| = |\mathbb{Z}^+|$?

odd negs.
even pos.
even neg. } bijective.

$$a_1, a_2, a_3, a_4, \dots$$

↓ ↓ ↓ ↓

$$O = \{1, 3, 5, 7, \dots\}$$

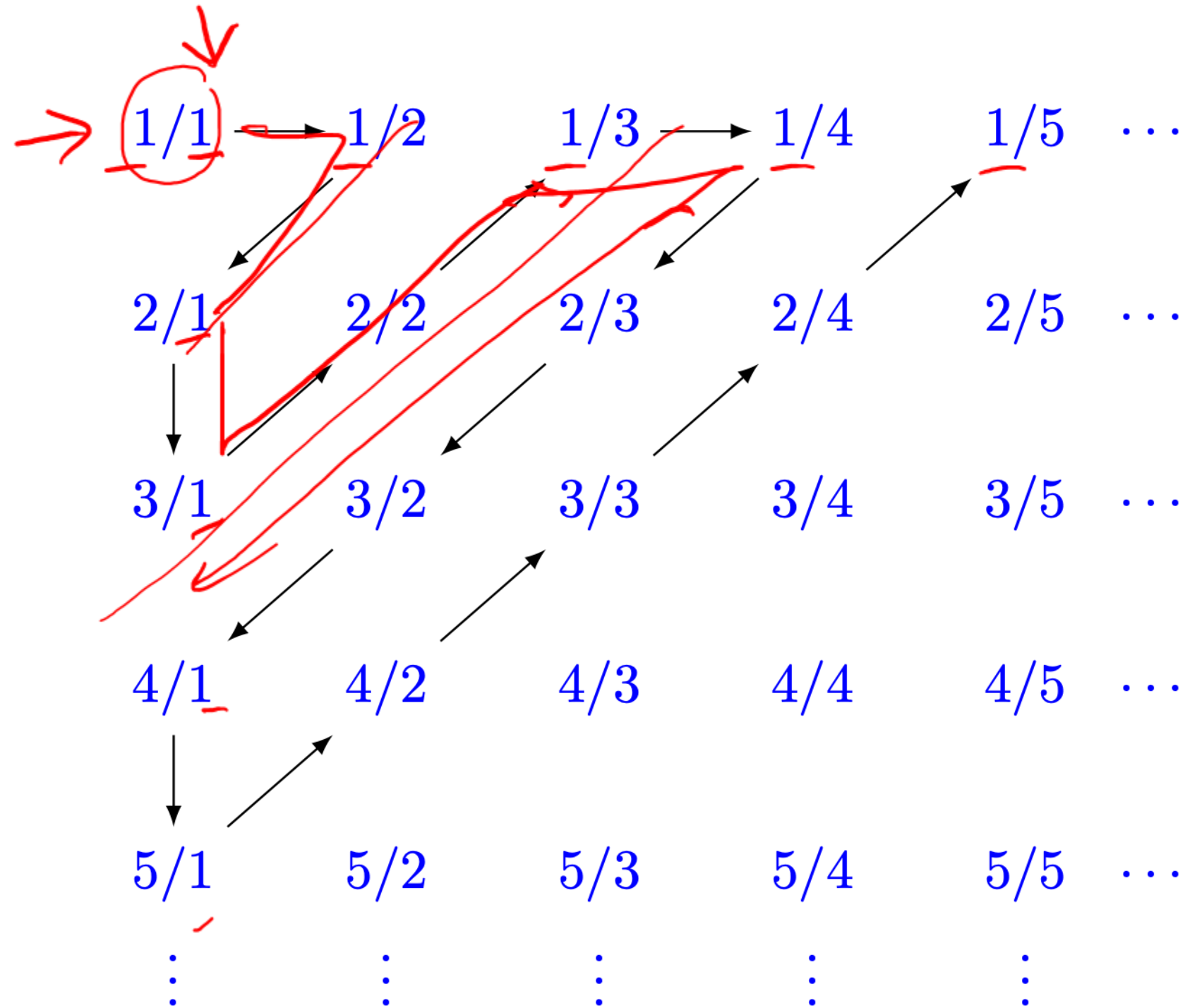
surjective:

$$o \in O, o = 2k - 1, k \in \mathbb{Z}^+ \\ f(k) = o, f \text{ surjective} \checkmark$$

Example

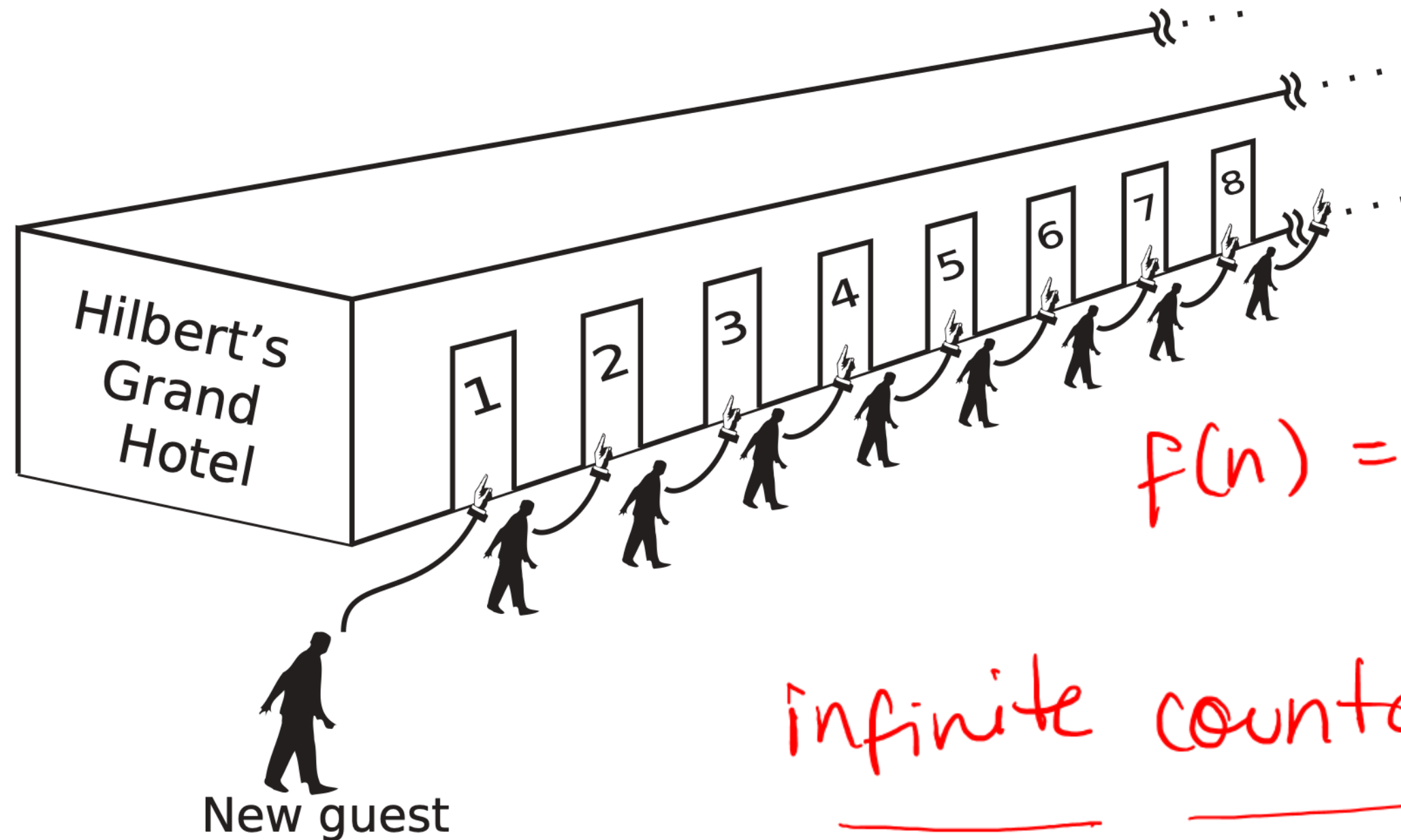
$$\mathbb{Q}^+ \rightarrow \mathbb{Z}^+ \rightarrow \{a_1, a_2, a_3, \dots, a_n, \dots\}$$

The set of positive rational numbers $\mathbb{Q}^+$ is countable.



Hilbert's Grand Hotel

How can we accommodate a new guest arriving at the fully occupied [Grand Hotel](#) without removing any of the current guests?



An Uncountable Set

Theorem The set of real numbers in $(0, 1)$ is uncountable.

Proof Suppose $(0, 1)$ is countable: $r_1, r_2, r_3, \dots$. Write

$$\begin{aligned} r_1 &= 0.\overline{d_{11}}\overline{d_{12}}\overline{d_{13}}\overline{d_{14}}\cdots \\ r_2 &= 0.d_{21}d_{22}d_{23}d_{24}\cdots \\ r_3 &= 0.d_{31}d_{32}d_{33}d_{34}\cdots \\ &\vdots \end{aligned}$$


Where $d_{ij} \in \{0, 1, 2, \dots, 9\}$. Build a new number $r = 0.\overline{d_1d_2d_3d_4} \dots$ such that

$$d_i = \begin{cases} 4 & \text{if } d_{ii} \neq 4 \\ \underline{5} & \text{if } \underline{d_{ii}} = 4. \end{cases}$$

Then $r \in (0, 1)$, but it is not in the list $r_1, r_2, r_3, \dots$, a contradiction.

Useful Results

- Any subset of a countable set is countable.

$$A = \{a_1, a_2, \dots, a_n, \dots\}$$

$$B \subseteq A, \quad B \text{ elements } a_i, a_{i+1}, a_{i+2}, \dots \in A$$

- If A and B are countable, then $A \cup B$ is also countable.

$$A = \{a_1, a_2, \dots\}, \quad B = \{b_1, b_2, \dots\}$$

$$A \cup B = \{a_1, b_1, a_2, b_2, \dots\}$$

$\uparrow \quad \uparrow \quad \uparrow$
 $c_1 \quad c_2 \quad c_3 \quad \dots$